Multi-Mode Smart Hand-Gesture Mouse

University of Sri Jayewardenepura

Faculty of Applied Sciences

ICT 305 2.0 Embedded System

Project Report

Multi-Mode Smart Hand-Gesture Mouse

By

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Acknowledgments

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1.Purpose of the Project

The primary purpose of developing this **Bluetooth Hand-Gesture Mouse** is to create a more intuitive, flexible, and portable human–computer interaction device that overcomes the limitations of traditional computer mice. Standard mice require flat surfaces, continuous hand movement, and physical scrolling mechanisms, which can be inconvenient during presentations, mobile usage, or when space is limited.

This project aims to introduce a **gesture-controlled, wireless mouse** that can be operated using simple hand movements and button inputs. By using an **MPU6050 gyroscope**, the device allows cursor control through natural hand tilting, enabling touch-free interaction. The addition of a **scroll function controlled by tilt gestures**, and a **laser presentation mode**, makes the device highly versatile for classrooms, meetings, and everyday computer use.

This enhances user experience and demonstrates efficient integration of hardware and software technologies.

Ultimately, the project showcases how embedded systems can be used to develop innovative, user-friendly input devices that provide greater freedom, portability, and modern functionality compared to conventional mice.

2. Introduction

This project presents a **Bluetooth-based Smart Hand-Gesture Mouse** designed using an **ESP32** microcontroller. Traditional mice use mechanical/optical sensors, but this system introduces:

- **Gesture-controlled movement** using **MPU6050 gyroscope**
- **Scroll control using hand tilt + middle button**
- **Laser pointer mode for presentations**
- **OLED display for battery %, date & time, Bluetooth status, and mode**
- **Automatic Bluetooth connection indicator with LED**
- **Rechargeable system using TP4056 + 1000 mAh Li-ion battery**

This device is compact, fully embedded, and provides a unique human-computer interaction experience.

3. Project Objectives

- Develop a **Bluetooth mouse** without mechanical movement.
- Implement **gesture-based cursor control** using MPU6050 gyro data.
- Integrate **dual-mode functionality**:
    - Mouse Mode
    - Laser Pointer Mode
- Display **real-time status information** on an OLED screen.
- Design a **fully portable**, battery-powered embedded system.
- Ensure reliable and stable BLE-HID communication.

4. System Overview

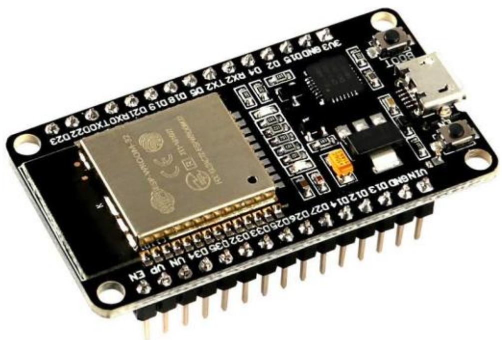
4.1 Main Features

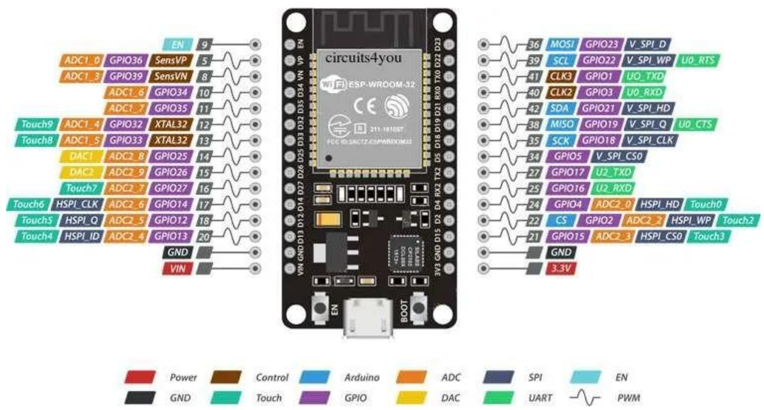
Feature	Description
Mouse Control	Gyro-based movement + left/right click
Scroll Function	Middle button + hand tilt (up/down)
Laser Mode	Laser active, mouse disabled
OLED Dashboard	Battery%, Date/Time, Mode, BLE status
Buttons	Left, Right, Middle, Mode toggle
Battery	1000 mAh Li-ion rechargeable
Charging	TP4056 with USB-C
LED Indicator	Connection/bonding status

5. Hardware Components

5.1 Microcontroller

- **ESP32 Dev Board**





- Built-in Bluetooth LE
- 2 cores, suitable for real-time tasks
- 3.3V operating voltage
- Sufficient GPIO pins for buttons & sensors

5.2 Sensors & Modules

- **5.2.1 Gyroscope & Accelerometer**

Used for:

- Hand angle detection
- Cursor movement
- Scroll direction

- **5.2.2 HW-111 RTC Module (DS3231)**

Used to display:

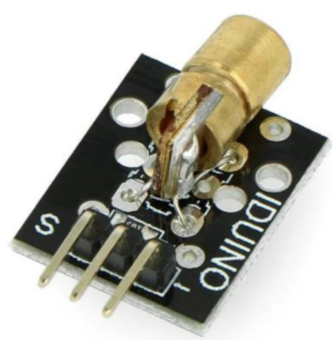
- Date
- Time
- 12/24-hour format

- **5.2.3 0.96" OLED Display**

Shows:

- Battery percentage
- Current time & date
- Bluetooth connection icon
- Mode indication (Laser / Mouse)

- **5.2.4 Laser Module**



Used in **Laser Mode** as a pointer for presentations.

- **5.2.5 LED Indicator**

- **Solid ON** → Device connected
- **Blinking** → Not connected

5.3 Power System

- **5.3.1 Li-ion Battery ()**

- Provides ~5–6 hours of usage
- 3.7V nominal voltage

- **5.3.2 Charger Module**

Used for:

- Charging the battery

- Power path for running while charging
- Protection against overcharge/over discharge

6. System Workflow

6.1 Button Functions

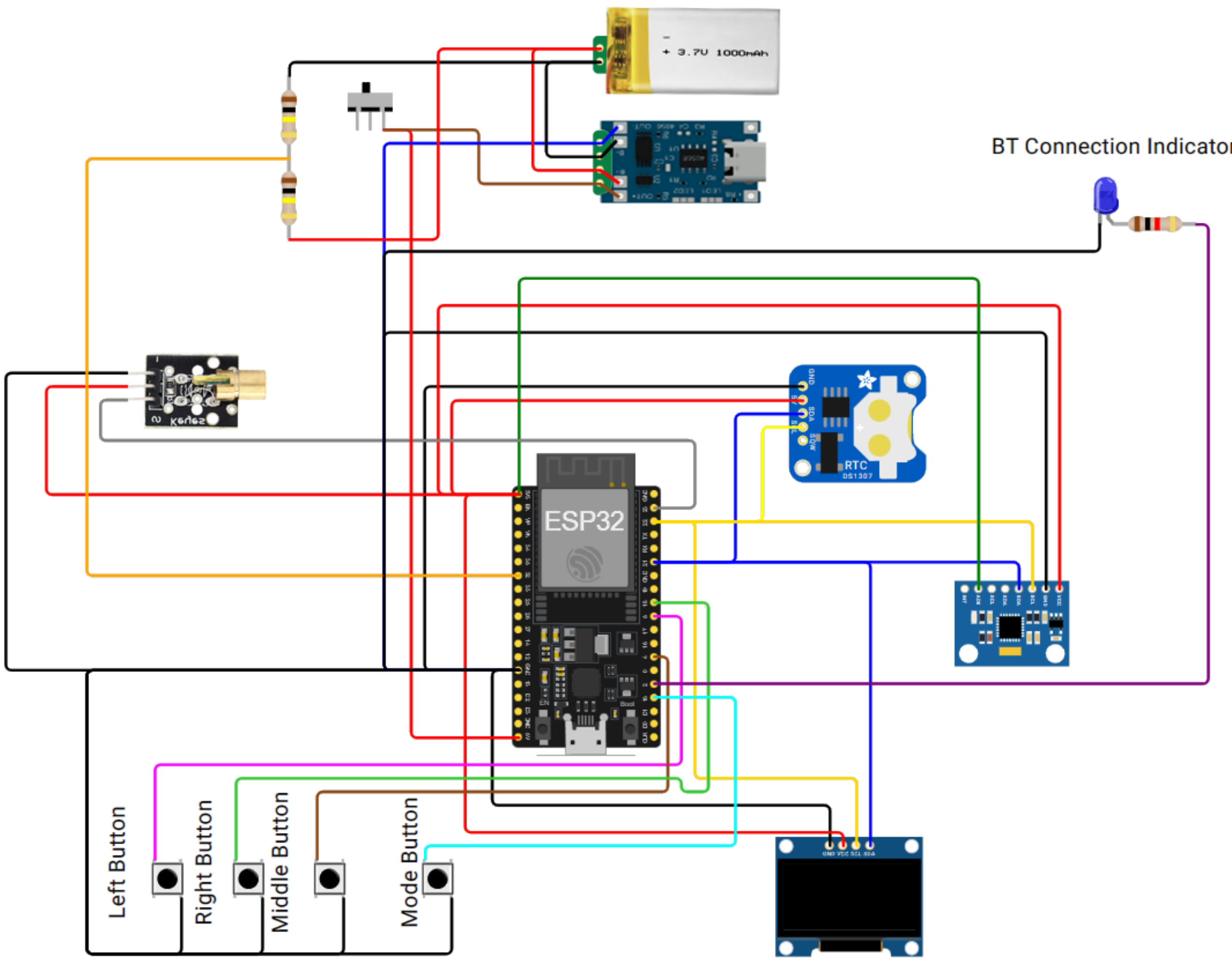
- **Left Button**
  - Works like normal mouse left click.
- **Right Button**
  - Works like normal right click.
- **Middle Button**
  - When **pressed**, cursor movement stops.
  - Tilting hand **up** → scroll up
  - Tilting **down** → scroll down
- **Mode Button**
  - Toggles between:
    - **Mouse Mode** (Laser OFF, mouse active)
    - **Laser Mode** (Laser ON, mouse disabled)

    A mode icon appears on the OLED screen.

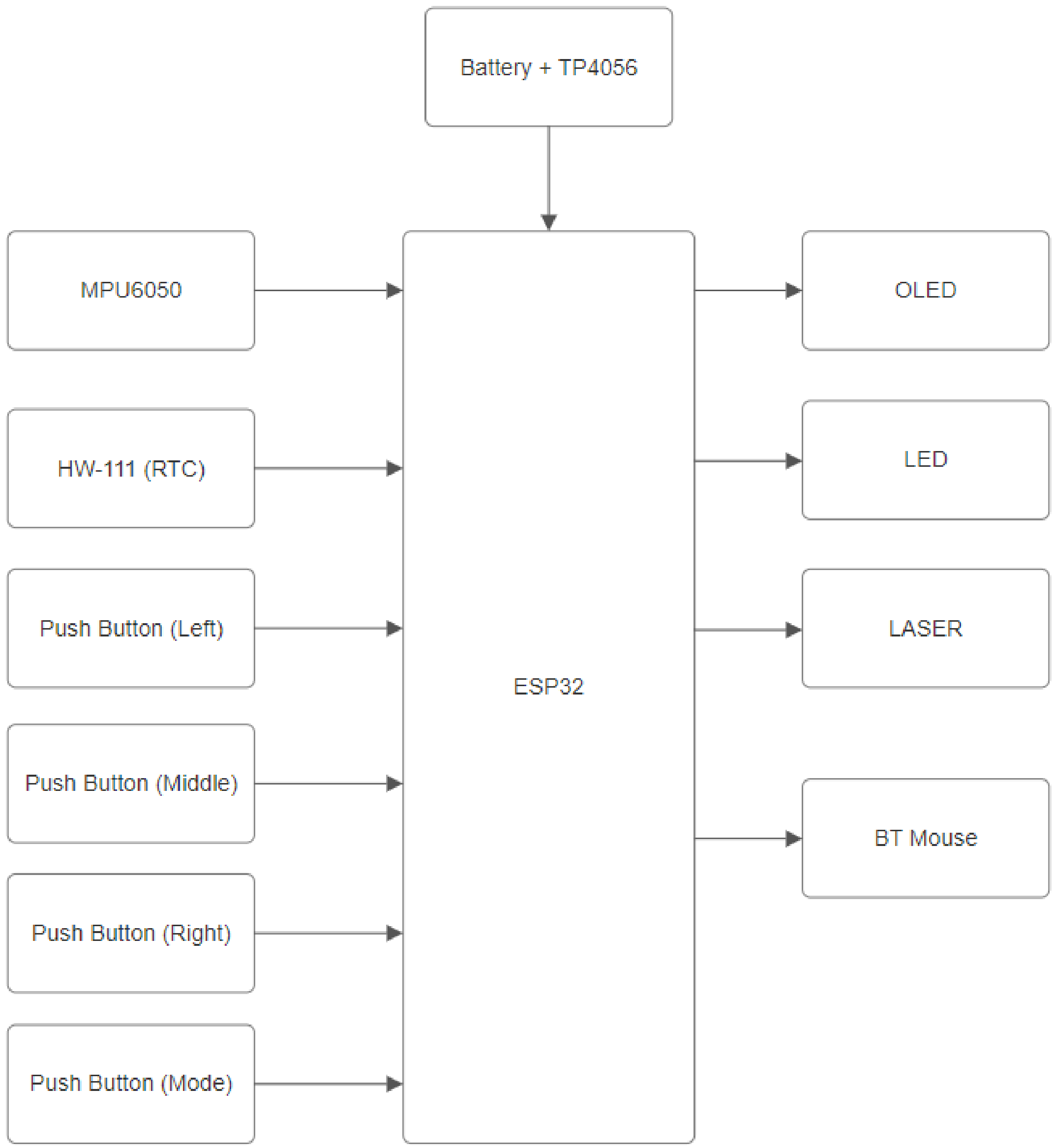
7. Power Management

- **TP4056** handles charging from USB-C
- ESP32 runs from **battery through** 3.3V regulator
- OLED shows **live battery %** using voltage divider & ADC
- While charging:
  - Device can operate
  - TP4056 LED indicators show charging statu**s**

8. Circuit Diagram



9. Block Diagram



10. Budget

Item	Quantity	Cost (Rs.)
• ESP32 Dev Board	1	1,250
• MPU6050	1	550
• HW-111 / DS3231 RTC	1	150
• 0.96" OLED Display	1	500
• Laser Module	1	110
• LED	1	5
• Push Buttons	4	40
• Switch	1	20
• Li-ion Battery (1000 mAh)	1	680
• TP4056 USB-C Charging Module	1	80
• Resistors (100k and 1k)	3	50
• PCB	1	90
• Gloves	1	290
• Wire	2 Sets	300
Total		4,115

11. Results & Performance

- Smooth cursor movement via gyro
- Scroll works accurately using tilt gestures
- Laser mode works perfectly for presentations
- OLED status dashboard is clear and readable
- Device works while charging
- Reliable Bluetooth HID connection with laptops and mobiles
- Battery lasts several hours per charge

12. Applications

- Human-Computer Interaction Devices
- Gesture-controlled interfaces
- Presentation remote
- Assistive technology
- Smart wearable controllers
- Gaming gesture input

13. Future Improvements

- Add haptic vibration feedback
- Use a more compact PCB design
- Add auto sleep mode to save battery
- Add sensitivity adjustments via menu
- Improve gesture recognition with IMU fusion filters

14. References

- ESP32 Technical Reference
- MPU6050 Datasheet
- TP4056 Charging Module Documentation
- Adafruit Libraries
- BLE HID Mouse Library

14.Discussion

The **Bluetooth Hand-Gesture Mouse** successfully integrates multiple embedded components into a portable and functional device. Cursor movement is controlled via the **MPU6050 gyroscope**, while the **middle button** allows gesture-based scrolling. The **mode button** toggles between **Mouse Mode** and **Laser Mode**, making the device versatile for presentations.

The **OLED display** provides real-time feedback on **battery level, date/time, Bluetooth status, and current mode**, improving usability. The **ESP32** handles BLE communication, allowing reliable connection with laptops and smartphones.

Challenges such as sensor noise and differentiating scroll vs. cursor movement were solved through filtering and careful calibration. Overall, the device is **compact, energy-efficient, and user-friendly**, demonstrating innovative human-computer interaction.

15.Conclusion

The **Bluetooth Hand-Gesture Mouse** is a compact and innovative embedded system that combines **gesture-based control, dual-mode functionality, and real-time feedback**. It successfully demonstrates how an **ESP32, MPU6050, OLED, and other modules** can work together to create a wireless, user-friendly device.

The project highlights improvements over traditional mice by enabling **scrolling through hand tilt, laser presentation mode, and portable operation**. Overall, it showcases a practical application of embedded systems in **modern human-computer interaction**, providing both functionality and convenience.